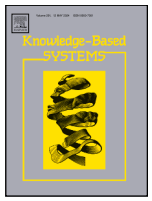
Knowledge-Based Systems 332 (2025) 114884



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys





Is there a better way to forget? Modelling memory decay in deep knowledge tracing

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ARTICLE INFO

Keywords:
Knowledge tracing
Educational data mining
Learning and forgetting
Memory decay
Deep learning

ABSTRACT

Knowledge acquiring and forgetting are integral components of the student learning process, both of which are critical for Deep Knowledge Tracing (DKT) tasks that leverage students' historical learning data to forecast future performance using deep learning methods. However, forgetting mechanisms proposed in many DKT models are not grounded in adequate theories and lack clear interpretation. Building on psychological theories, this study focuses on a crucial aspect of modelling forgetting - memory decay - and performs ablation study and comparative analysis of various decay functions in DKT models to investigate the robustness of their implementation of forgetting. Rather than proposing a new model, our goal is to understand how forgetting is manifested in the popular DKT task, identify shortcomings, and highlight potential opportunities to inform the design of new models grounded in forgetting-related theories. Our results show that the wide-adopted Ebbinghaus's forgetting curve underperforms when replaced by sigmoid and inverse decay functions in some scenarios. We also present a discussion on the factors that influence the models' performance, including the effects of completely removing forgetting components from DKT models and their computational implications.

1. Introduction

Deep Knowledge Tracing (DKT) is a deep learning task that enhances personalised education. It predicts students' future performance and identifies areas for improvement by modelling their mastery of skills over time [1]. In Knowledge Tracing (KT) tasks (see Fig. 1), learning and forgetting are interconnected in the dynamics of knowledge acquisition and decay. This is critical for creating accurate predictive models of student performance: while learning increases the likelihood of correctly answering future questions, forgetting decreases this likelihood over time.

Recent DKT studies that integrate advanced neural network architectures not only model students' knowledge acquisition process but also incorporate a decay function as the forgetting component to more accurately reflect the learning processes of students [2–4].

Thorndike [5] posited that all memory decays, akin to a muscle that will atrophy if not exercised. Memory decay plays a crucial role in the forgetting process, acting as a natural process where memory traces weaken over time if not activated or rehearsed [6]. This concept is grounded in the theory that the neural connections forming a memory become weaker, making the retrieval of information more difficult [7]. Ebbinghaus [8] suggested that immediately after learning, memory retention is at its highest, but it rapidly decreases exponentially unless the information is consciously reviewed. Following this suggestion, the DKT models that considered the time-based decay have designed an exponentially decaying kernel function to simulate forgetting behaviour consistently [2–4,9,10].

However, not all memories decay in the same way [11,12]. Psychological studies have commonly classified memories based on the lengths of time they have been stored (e.g., short-term and long-term memories) [13] and the type of information they relate to (e.g., semantic and perceptual memories) [14], proposing that different memory types should follow different decay curves [6]. This challenges Ebbinghaus's "one size fits all" forgetting curve and underscores a significant issue with the exponential decay function used in SOTA DKT models, which lack robust theoretical support.

Thus, this study aims to address the research question: *Is there a better way to forget?*, by analysing memory decay, an important aspect of forgetting, in DKT tasks. This study identifies and raises awareness of a fundamental problem in current DKT implementations: the overreliance on a single decay function without consideration of memory type diversity. Rather than proposing new model architectures, our goal is to expose shortcomings in existing approaches and highlight opportunities for theoretically grounded research in a field with over 13,000

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https://doi.org/10.1016/j.knosys.2025.114884
Received 9 April 2025; Received in revised form 20 July 2025; Accepted 9 November 2025
Available online 13 November 2025
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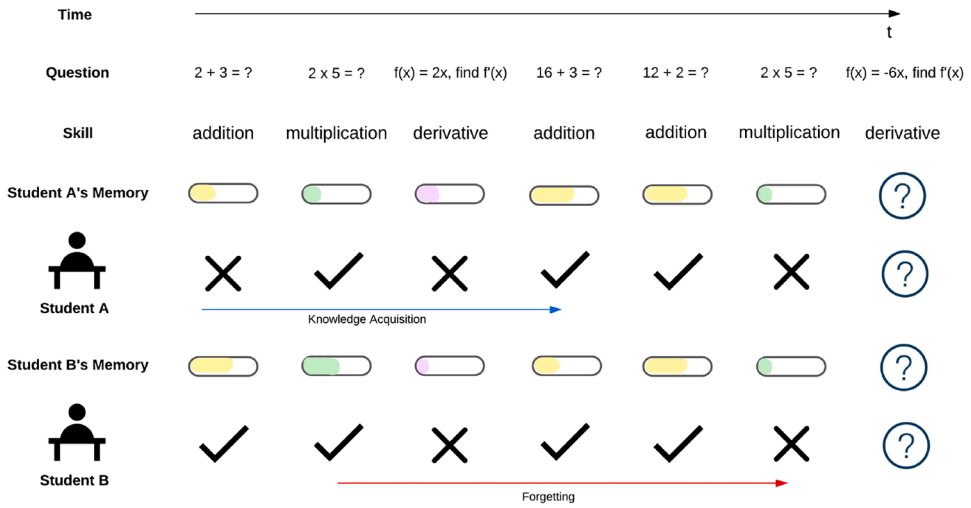


Fig. 1. A graphical representation of knowledge acquisition and forgetting in the KT problem.

conference papers and approximately 5000 journal articles published only in the ACM Digital Library[1]. For this, we conduct an ablation study and a comparative analysis of decay functions in several SOTA DKT models to uncover the dynamics of memory decay by assessing the impact of different forgetting curves on the performance of DKT models. Additionally, we provide comprehensive computational efficiency analysis to examine the trade-offs between model performance and computational cost, enabling informed practical decisions for researchers and practitioners developing DKT systems.

2. Related work

2.1. Forgetting in knowledge tracing

Existing KT models can be broadly categorised into traditional models and deep learning models. Traditional KT models rely on probabilistic models (e.g., Bayesian models) or statistical methods (e.g., Item Response Theory) to trace a student's changing knowledge states, while DKT models are inspired by recent advances in deep learning. DKT was first introduced by Piech et al., by incorporating Recurrent Neural Networks (RNNs), which includes Long Short Term Memory (LSTM) networks, to capture complex representations of student knowledge [1]. IEKT uses the Gated Recurrent Unit (GRU) variant to enhance prediction [15]. CKT uses Convolutional Neural Networks (CNNs) to model individual prior knowledge and learning rates of students [16]. Inspired by CKT, GKT reconstructs the KT task into a time-series node-level classification problem using Graph Neural Networks (GNNs) [17]. Recently, Transformer architecture and self-attention mechanisms have been employed in models such as SAKT [18] and SAINT [19] to improve performance by addressing data sparsity issues.

Forgetting, a crucial aspect of student learning, has been integrated into DKT models, enhancing their ability to detect knowledge deterioration through incorrect answers. "DKT-forget" improves the DKT by considering the practice interval and attempts for each question [20]. LPKT [21] utilises these intervals to better represent the impact of forgetting on learning. Similarly, SAINT+ [22] builds on SAINT [19] by including not only the learning intervals but also response time to model forgetting. Overall, DKT models with forgetting mechanisms generally outperform those without them, but the degree of improvement varies across datasets and warrants further investigation, which our current work aims to address.

[1] https://dl.acm.org

2.2. Forgetting approaches in DKT

Forgetting approaches have been widely modelled in recent DKT studies. DKVMN [23] is one of the first DKT models to simulate forgetting by updating a memory matrix that adjusts students' knowledge states with each answer. Successor models like GKT [17], Deep-IRT [24], and SKVMN [4] adopted similar mechanisms. DGMN [25] utilises an erase function in its attention memory for this purpose. LPKT [21] and LFBKT [26] calculate forgetting by multiplying a forgetting matrix with the knowledge embedding matrix to simulate the loss of knowledge.

Other models apply time-based decay functions to parameters to reflect forgetting. KPT [27] uses exponential decay to represent forgetting. DAS3H [28] weighs recent interactions more heavily than older ones to estimate skill mastery. AKT uses a monotonic attention mechanism that takes the basic form of an exponential decay curve over time to incorporate knowledge decay [4]. HawkesKT [3] and RKT [2] use time gaps and a decaying kernel function, respectively, to lessen the impact of past interactions over time. NGFKT [10] and CAKT [9] integrate exponential functions related to time in their computations, while FoLiBi [29] introduces a forgetting-aware bias to capture knowledge decay. Models that use time-based decay have a common feature of integrating the Ebbinghaus exponential forgetting curve. This curve is applied as a mathematical function to model the reduction of memory retention over time. Our work challenges the use of the Ebbinghaus curve as a one-size-fits-all solution and tests other decay curves to determine how to better represent forgetting.

2.3. Memory decay

Forgetting is often characterised by the inability to express previously acquired memory in situations where it would normally be recalled. Psychological studies have identified memory decay as a primary process in forgetting, which refers to the gradual fading of stored memories [6,11]. Ebbinghaus's forgetting curve [8], as one of the earliest and most influential theories of memory decay, describes the exponential loss of information over time unless it is revisited, which has been widely adopted by DKT models [2–4,9,10].

However, contemporary cognitive psychology research reveals that memory comprises fundamentally different types with distinct neurobiological underpinnings and decay characteristics [30,31]. Semantic memory, responsible for factual knowledge and concepts, tends to follow exponential decay patterns consistent with Ebbinghaus's curve, particularly when information lacks contextual reinforcement [32]. In contrast, procedural memory, which governs skill-based learning and

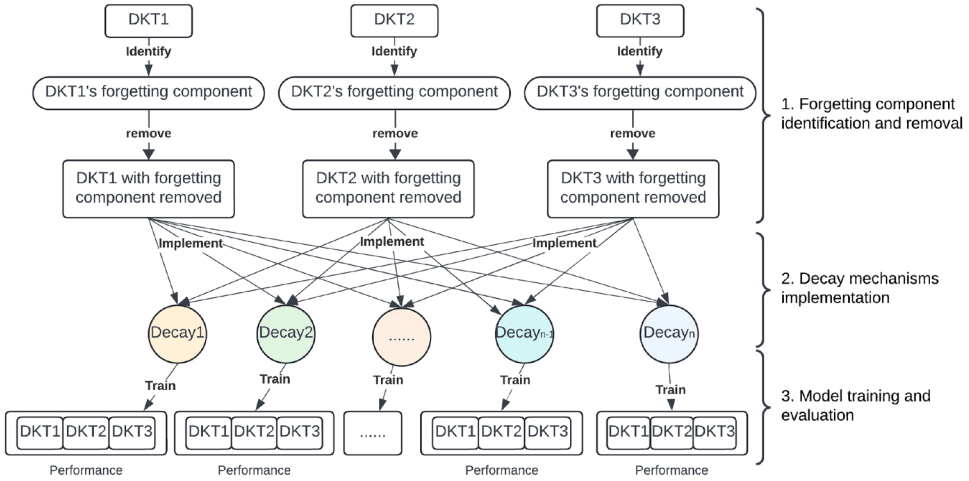


Fig. 2. Our three-step methodology overview.

motor sequences, exhibits more gradual, power-law decay patterns that align with polynomial or inverse functions [33]. This distinction is crucial for educational contexts where students acquire both declarative knowledge (facts, concepts) and procedural skills (problem-solving techniques, mathematical operations).

Episodic memory, which encodes specific learning events and their temporal-spatial context, demonstrates sigmoid-like decay patterns where initial retention remains high before experiencing rapid decline, followed by stabilisation [34,35]. This pattern reflects the consolidation process where memories transition from hippocampal-dependent storage to neocortical long-term storage [36]. Working memory, fundamental to online learning tasks, shows inverse decay characteristics where information degrades rapidly unless actively maintained through rehearsal [37]. These theoretical distinctions suggest that educational systems involving multiple memory types require correspondingly diverse decay functions rather than universal exponential decay.

3. Methodology

To understand the impact of forgetting components on DKT models, we propose the three-step methodology shown in Fig. 2: Firstly, 1) identifying the forgetting component within the model's architecture and removing it to analyse its effect on model performance. We then 2) implement various decay functions to explore the influence of memory decay mechanisms in DKT models. Lastly, we 3) retrain each model with different implementations and evaluate the performance using measures including Analysis of Variance (ANOVA), accuracy (ACC), the area under the receiver operating characteristic curve (AUC) and the root mean squared error (RMSE).

3.1. Forgetting component identification and removal

KT predicts whether a student can answer the next question based on their previous interactions. Each interaction $x_t = (q_t, c_t, r_t)$ consists of a tuple of a question q_t, the knowledge component (KC) c_t which is either a skill or a concept, and its correctness $r_t \in \{0, 1\}$. A knowledge state represents a student's current level of mastery and understanding of specific KCs over time, denoted as h_t, which considers x_t. The goal is to predict a student's knowledge state h_T and the correctness of the student's next answer r_T, based on their prior sequence of interactions X_T ($X = <x_1, x_2, \ldots, x_{T-1}>$). DKT models typically predict a student's knowledge state across various concepts over time, based on their interaction history with learning materials. The forgetting function F in DKT modifies the influence of past interactions based on their distance from the current prediction point in a model:

$$F_{model} = \exp(-\theta \cdot d(t, \tau))$$

where $\theta > 0$ is a learnable decay rate parameter and $d(t, \tau)$ is the temporal distance between the time step of a particular question t and the time step of a past question τ.

After identifying the forgetting component F used by each model, we then remove it to assess its effectiveness on model performance. We perform a generalised modification across DKT models by neutralising this forgetting effect, setting F to uniformly treat all interactions irrespective of their distance. This is achieved by either removing F from the model's computations or setting it to a constant value of 1, particularly if the decay function F is multiplied by another variable. We then reassess the model's performance after removing the decay function to measure the effect of this modification.

3.2. Decay mechanisms implementation

This step explores the integration of different decay mechanisms (see Table 1) in DKT models to assess their impact on model performance, as shown in Fig. 2. This integration involves the implementation of logarithmic, sigmoid, inverse, step, linear, and polynomial decay functions as replacements for the classical exponential decay function.

We acknowledge that DKT models integrate these decay functions at different architectural points-attention mechanisms in transformer-based models, memory matrices in memory-augmented networks, and cell states in recurrent architectures. Rather than normalising these integration points, we deliberately preserve these architectural differences to investigate how the same decay function behaves across diverse model structures. This approach reflects real-world deployment scenarios where practitioners often choose appropriate decay functions for existing model architectures rather than redesigning models from scratch.

Table 1
Overview of Decay Functions.

Name	Function	Parameters	Theory
Exponential	$e^{-\mu d}$	μ: decay rate; d: distance	Ebbinghaus [8]
Logarithmic	$\frac{1}{\log(\alpha d+e)}$	α: rate modifier	Jost [38,39]
Sigmoid	$\frac{1}{1+e^{-\beta(d-\delta)}}$	β, δ: shape params	S-learning curves [40]
Inverse	$\frac{1}{\gamma d+1}$	γ: inverse rate	Diminishing Returns [33]
Step	$\begin{cases} 1, & d < \theta \\ 0, & \text{else} \end{cases}$	θ: threshold	Threshold theory [41]
Polynomial	$\frac{1}{(\kappa d+1)^n}$	κ, n: rate, order	Crossman's Law [42]

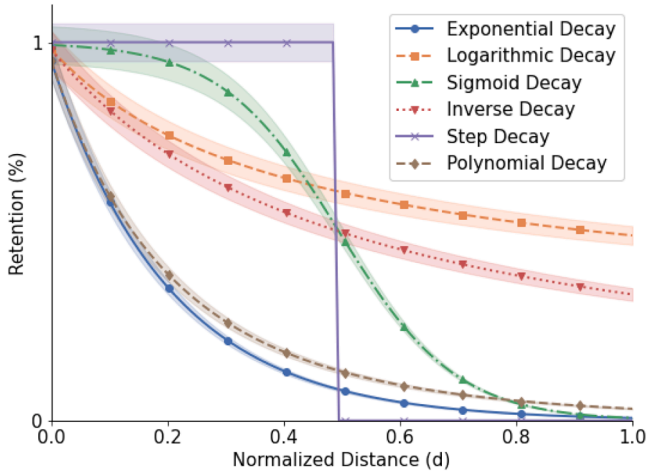


Fig. 3. General shapes of different decay curves.

Each decay function is characterised by specific parameters designed to modulate the rate and nature of knowledge decay over time. Crucially, these mathematical formulations correspond to distinct memory types identified in cognitive psychology, enabling principled selection based on learning content and context. The default shapes are demonstrated in Fig. 3. Here, the exponential decay coming from Ebbinghaus's forgetting curve is treated as the benchmark. It illustrates that memory retention declines exponentially over time without reinforcement, showing a rapid initial loss of information, which gradually stabilises as time passes. This pattern aligns with semantic memory decay, where factual knowledge and concepts deteriorate rapidly without contextual reinforcement, particularly in isolated learning environments [43,44].

The logarithmic decay introduces a gradual reduction in influence with increasing distance which aligns with Jost's observation that older memories decay more gradually [38,39]. This slower decay pattern corresponds to consolidated long-term memories that have undergone repeated rehearsal and integration into existing knowledge structures [45,46].

In contrast to the exponential decay function, sigmoid decay has a smoother drop in the initial high decay rate [35]. It is derived from the S-shaped learning curve [40], which illustrates that learning initially progresses slowly as individuals familiarise themselves with new material. This is followed by a rapid improvement phase as proficiency increases, and finally, a plateau where further learning slows down. Significantly, sigmoid decay patterns mirror episodic memory consolidation processes, where initial encoding maintains high retention before experiencing rapid decay during the consolidation window, ultimately stabilising in long-term storage [47,48]. This pattern is particularly relevant for contextual learning events and problem-solving episodes in educational settings.

Inverse decay is more commonly discussed in the law of diminishing returns where the initial periods of practice yield significant improvements, but over time, additional practice results in smaller gains [33]. This decay pattern closely models working memory dynamics, where information degrades rapidly unless actively maintained through rehearsal mechanisms [49]. Recent research on subliminal memory exposures demonstrates inverse forgetting patterns, suggesting that initially weak memories may be prioritised for consolidation, resulting in the observed inverse decay characteristics [50].

Step decay models a sudden change in decay rate at a specific point, aligning with threshold theory which posits that responses or effects occur only when a certain threshold is surpassed [41]. This discrete transition reflects interference theory principles, where competing knowledge items suddenly displace existing memories once critical interference thresholds are exceeded [51].

Polynomial function is grounded in Crossman's Law which suggests that the time taken to halve the error rate in a repetitive task remains constant [42]. This power-law relationship characterises procedural memory decay, where motor skills and cognitive procedures follow predictable practice-dependent retention patterns [52]. Parameter n, referred to as the "order" of the polynomial, determines the degree of the polynomial in the denominator of the fraction, influencing how the function behaves as d changes. According to the polynomial degree, polynomial decay curve typically starts with a sharp decrease or increase, contains several inflection points where the direction of the curve changes, and then gradually levels off, approaching the axis as it extends.

3.3. Model training and evaluation

The operationalisation of these decay mechanisms requires the careful tuning of their respective hyper-parameters (α, β, γ, θ, μ, κ, and n) to optimise model performance. This parameter tuning process is executed through cross-validation on five mostly-used KT datasets. After the training step, we evaluate the impact of removing the forgetting component and implementing each decay function on the model's performance. Performance metrics including ACC, AUC, and RMSE serve as the basis for our evaluation.

To evaluate the efficacy of various decay functions within DKT models, we further perform an ANOVA test to assess the significance and impact of different decay mechanism implementations on model performance.

4. Experiment

4.1. Datasets selection

As listed in Table 2, five commonly-used KT datasets, AS12 [53], AS17 [54], BA06 [55], NIPS34 [56], EN1 [57], are selected to train and evaluate DKT models with different implementations of decay mechanisms. The primary reason for selecting these educational datasets is their inclusion of timestamps related to students' learning activities, as they are crucial for models that account for forgetting [58]. Datasets such as ASSISTments2015 and Statics2011, which lack this attribute, were not included in our analysis. Additionally, to assess the robustness of DKT models against various feature implementations, it is essential that the datasets vary in their characteristics and purposes. A detailed description of the datasets is as below:

Table 2
Statistics of the Selected Datasets.

	AS12	AS17	BA06	NIPS34	EN1
# of questions	45,716	3162	207,856	948	13,169
# of KCs	265	102	493	57	188
# of students	27,066	1709	1146	4918	784,309
# of answers	2,541,201	942,816	3,656,871	1,382,727	95,293,926
Timestamp	✓	✓	✓	✓	✓
Purpose	Educational research	Longitudinal prediction	Mathematics-focused prediction	Diagnostic questions	AI tutoring
Questions Bundling	✓	✓			✓

Table 3
Summary of the selected DKT models.

Model	Technique	Forgetting Implementation
AKT*, SimpleKT*	Rasch model	Modification on the scaled dot-product attention mechanism
HawkesKT, CT-NCM	Hawkes process	Introduction of a fine-grained temporal cross effects term
RKT, DKT-Forget*	RNN	Introduction of a forgetting-based relation vector

* refers to SOTA models.

ASSISTments2012 (AS12)[2] is collected from skill builder problem sets (2012–2013) on ASSISTments, an online tutoring system. Students solved similar problems aligned with the same KCs, with the dataset providing attributes on student behavior and unique identifiers for teachers and schools.

ASSISTments2017 (AS17)[3] comes from the ASSISTments Longitudinal Data Mining Competition (2017), aiming to predict students' likelihood of securing STEM jobs after college. It contains a decade of student learning activities on the platform.

Bridge to Algebra 2006–2007 (BA06)[4] derives from the KDD Cup 2010 EDM Challenge, containing data from 13 to 14 year-old students using "Bridge to Algebra" tutoring materials (2007–2008). It focuses on mathematical questions with multiple KCs.

NeurIPS2020 Education Challenge (NIPS34)[5] comes from the NeurIPS 2020 Education Challenge, featuring multiple-choice programming questions designed to test students' understanding of misconceptions.

EdNet-KT1 (EN1)[6] is collected from Santa[7], an AI tutoring platform, containing various student computing activities from question-answering to lecture consumption.

4.2. Model tuning

Six recent and popular DKT models (including the SOTA models) following different model architectures and incorporating forgetting components were selected for our comparative experiments. They are: AKT [4], SimpleKT [59], HawkesKT [3], CT-NCM [60], DKT-forget [20] and RKT [2]. The diversity of architectures is crucial for evaluating the generalisability and robustness of our experiments. Table 3 summarises the selected models and their common features. As each pair of models shares a similar architecture for modelling memory decay, we focus on one model from each group to detail the model-tuning approach.

4.2.1. AKT Model

AKT implements forgetting by integrating an exponential decay term F_{AKT} into its monotonic attention mechanism. F_{AKT} is formulated as $\exp(-\theta \cdot d(t,\tau))$ (as explained in Step 1 of Methodology section), which involves using the exponential decay function to down-weight the importance of questions from the distant past. The integration of F_{AKT} into AKT's monotonic attention mechanism is formulated as:

$$\alpha_{t,\tau} = Softmax\left(\frac{F_{AKT} \cdot \mathbf{q}_t^T \mathbf{k}_\tau}{\sqrt{D_k}}\right) \tag{1}$$

[2] https://sites.google.com/site/assistmentsdata/2012-13-school-data-with-affect
[3] https://sites.google.com/view/assistmentsdatamining/
[4] https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp
[5] https://eedi.com/projects/neurips-education-challenge
[6] http://ednet-leaderboard.s3-website-ap-northeast-1.amazonaws.com
[7] https://ai.aitutorsanta.com/

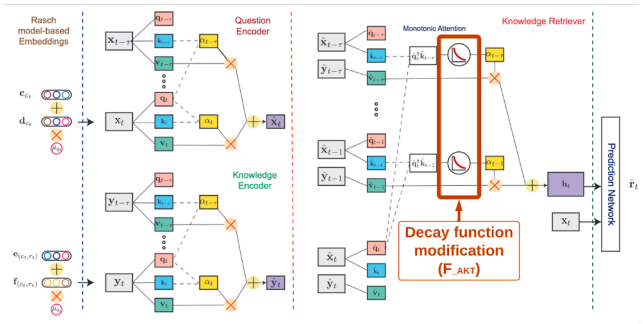


Fig. 4. Modification of decay function in AKT's model architecture (original image is from [4]).

where $\mathbf{q}_t$ is the query corresponding to a particular question, $\mathbf{k}_\tau$ is the key corresponding to a past question q_τ, and D_k is the dimension of a respective key embedding in the AKT model.

To assess the impact of F_{AKT}, we exclude it from Eq. 1, resulting in a modified equation as:

$$\alpha_{t,\tau} = Softmax\left(\frac{\mathbf{q}_t^T \mathbf{k}_\tau}{\sqrt{D_k}}\right) \tag{2}$$

F_{AKT} is then modified to replace the exp() decay function with other decay functions introduced in Table 1. The location where the exp() decay function is replaced within AKT's architecture is shown in Fig. 4. For example, Eq. 3 shows the replacement of exp() decay function with the Logarithmic decay in $FAKT$:

$$\alpha_{t,\tau} = Softmax\left(\frac{\mathbf{q}_t^T \mathbf{k}_\tau}{log(\theta d(t,\tau)+e)\sqrt{D_k}}\right) \tag{3}$$

Subsequently, we obtain the optimal hyperparameters for each decay function to replace the exp() decay function with the inverse function, step function with $\theta = 0.5$, polynomial function with $n = 5$ and sigmoid function respectively.

4.2.2. Hawkeskt model

This model incorporates forgetting as a kernel function that controls the time-sensitive impacts of past interaction x_t on the current interaction x_T at a fine-grained level. The kernel function is formulated as:

$$F_{HawkesKT}(T-t) = exp(-(1+\beta_{x_t,x_T})log(T-t)) \tag{4}$$

where β_{x_t,x_T} is a learnable parameter controlling fine-grained decay rates under different circumstances. $F_{HawkesKT}(T-t)$ uses the $exp()$ decay function to model memory decay at different time steps. It is then integrated into HawkesKT's intensity function [61], which directly predicts r_T, formulated as:

$$r_T = \lambda_0^{x_T} + \sum_{x_t \in S_{T-1}} \alpha_{t,T} \cdot F_{HawkesKT}(T-t) \tag{5}$$

where $\lambda_0^{x_T}$ represents the inherent properties of x_T, S_{T-1} is the set of past interactions and $\alpha_{t,T}$ controls the time-sensitive impacts of every past interaction x_t on the current interaction x_T at an immediate level.

Forgetting is removed by excluding $F_{HawkesKT}(T-t)$ in Eq. 5, whilst alternative decay functions can be substituted. For example, sigmoid decay:

$$r_T = \lambda_0^{x_T} + \sum_{x_t \in S_{T-1}} \alpha_{t,T} \tag{6}$$

$$F_{HawkesKT}(T-t) = \frac{1}{1+e^{-(1+\beta_{x_t,x_T})log(T-t)+\delta}} \tag{7}$$

4.2.3. RKT Model

This model incorporates forgetting as part of the contextual information used to formulate the impact of past questions on the current question, which is used to predict r_T [2]. Given the time sequence of a

student's history of questions answering $t = (t_1, t_2, \ldots, t_{n-1})$, forgetting is formulated as:

$$F_{RKT} = \left[\exp\left(-\frac{t_n - t_1}{S_u}\right), \exp\left(-\frac{t_n - t_2}{S_u}\right), \ldots, \exp\left(-\frac{t_n - t_{n-1}}{S_u}\right)\right] \tag{8}$$

where S_u is a learnable parameter representing the relative strength of memory with respect to student u. Each element specifies the importance of the ith question with the current question. The more remote the ith question, the less relevant it is to the current question. Therefore, the exponential decay function is applied on the time difference between the ith question and the current question to reflect forgetting.

F_{RKT} is then added with a vector R^E, which represents the relationships between questions, to produce the ultimate relation vector R. This resultant vector R is used in calculating attention scores. R is replaced by R^E to remove forgetting effects, whilst alternative decay functions can be implemented by reformulating:

$$R^T = \left[f\left(-\frac{t_n - t_1}{S_u}\right), f\left(-\frac{t_n - t_2}{S_u}\right), \ldots, f\left(-\frac{t_n - t_{n-1}}{S_u}\right)\right] \tag{9}$$

where f is any function in Table 1 so that the robustness of $FRKT$ can be assessed.

4.2.4. Simplekt model

Memory decay is not considered in SimpleKT. Inspired by how decay has been modelled in AKT, we added a similar decay function that is modulated by a parameter γ to assess the impact of memory decay, which adjusts the influence of past interactions based on their temporal distance. Specifically, $F_{SimpleKT}$ is applied to the attention scores to down-weight the importance of distant past interactions before the softmax operation. The integration of $F_{SimpleKT}$ into SimpleKT's attention mechanism is formulated as:

$$s'_{t,\tau} = \frac{\mathbf{q}_t^\top \mathbf{k}_\tau}{\sqrt{d_k}} \cdot F_{\text{SimpleKT}}(\gamma, |t - \tau|) \tag{10}$$

where $\mathbf{q}_t$ is the query vector at time t, $\mathbf{k}_\tau$ is the key vector at time τ, d_k is the dimension of the key vectors, and $|t - \tau|$ represents the temporal distance between interactions. F_{SimpleKT} is modified to fit different decay functions as we summarised before.

4.2.5. CT-NCM

CT-NCM incorporates forgetting by integrating a decay function $F_{\text{CT-NCM}}$ into its continuous-time LSTM architecture. This decay function adjusts the cell state $\mathbf{c}_t$ over time based on the time lag Δt between interactions. Specifically, $F_{\text{CT-NCM}}$ modulates the influence of previous hidden states by applying a decay to the cell state $\mathbf{c}_t$ as:

$$\mathbf{c}_{\text{delay}} = \mathbf{c}_{\text{bar}} + (\mathbf{c} - \mathbf{c}_{\text{bar}}) \cdot F_{\text{CT-NCM}}(-\boldsymbol{\delta} \cdot \Delta t) \tag{11}$$

where $\mathbf{c}_{\text{bar}}$ is the baseline cell state, $\boldsymbol{\delta}$ is a learnable delay parameter, and Δt is the time lag since the last interaction. The exponential decay term $\exp(-\boldsymbol{\delta} \cdot \Delta t)$ is implemented in the original model as the decay function to reduce the influence of the previous cell state $\mathbf{c}$ as the time lag Δt increases. To remove forgetting, we set $\mathbf{c}_{\text{delay}} = \mathbf{c}_{\text{bar}}$. Alternative decay functions replace the exponential term, such as logarithmic decay:

$$\mathbf{c}_{\text{delay}} = \mathbf{c}_{\text{bar}} + (\mathbf{c} - \mathbf{c}_{\text{bar}}) \cdot \log\left(1 + \text{clamp}(-\boldsymbol{\delta} \cdot \Delta t)\right) \tag{12}$$

where clamp$(\cdot)$ ensures the input to the logarithm is within a valid range.

4.2.6. DKT-Forget Model

DKT + Forget models forgetting by integrating a multi-hot forgetting vector c_t into its input space and c_{t+1} into its output space respectively, which does not consider the effect of memory decay. We modify the model by integrating a decay function $F_{\text{DKT-Forget}}$ into its integration component, which adjusts the input embeddings based on temporal features including response gap, student gap, and practice count [20]. Specifically, $F_{\text{DKT-Forget}}$ modulates the influence of past interactions by applying a decay to the input embeddings $\mathbf{v}_t$ through element-wise multiplication, formulated as:

$$\theta_t = \mathbf{v}_t \odot F_{\text{DKT-Forget}}(C\mathbf{c}_t) \tag{13}$$

where $\mathbf{v}_t$ is the input embedding at time t, C is a linear transformation matrix, $\mathbf{c}_t$ is the concatenated time features, and $\odot$ denotes element-wise multiplication. Alternative decay functions replace $F_{\text{DKT-Forget}}$, such as logarithmic decay: $\theta_t = \mathbf{v}_t \odot \log(1 + |C\mathbf{c}_t|)$.

Table 4
ANOVA analysis for modifies in DKT models.

Model	Factor	SS	df	F	Sig.
RKT	Intercept	43.66	1	5646.43	<.05
	Method	0.000004	6	0.856	>.05
AKT	Intercept	60.93	1	5638.22	<.05
	Method	0.075	6	10.10	<.05*
HawkesKT	Intercept	40.28	1	4340.90	<.05
	Method	0.007	6	4.714	<.05*
SimpleKT	Intercept	44.81	1	1096.19	<.05
	Method	0.007	6	7.941	<.05*
CT-NCM	Intercept	40.941	1	1380.01	<.05
	Method	0.000007	6	0.0047	>.05
DKT-Forget	Intercept	42.454	1	10173.06	<.05
	Method	0.001	6	5.349	<.05*

* p<0.05 significant. df: degrees of freedom; SS: Sum of Squares; F: F-Statistic.

4.3. Experimental setup

We use EduStudio[8] - an unified library for KT based on Pytorch, to ensure consistent and reproducible comparisons of model performance. In EduStudio's data preprocessing pipeline, raw datasets are first imported from their original sources. These datasets are then transformed through model-specific atomic operations into a middle data layer. A set of atomic operations standardises the middle data into cache data with consistent format, ready for model training [9]. In our experiments, we configured the decay function by adjusting the *decay_function* parameter in the model's configuration file. Specifically, we set it to *exp* (exponential decay) for models like AKT, HawkesKT, RKT, and CT-NCM, as per their original implementations, and to *non* for DKT-Forget and SimpleKT, which do not originally incorporate a memory decay function. The detailed implemention steps and our code are documented at https://anonymous.4open.science/r/DKT-Decay-Implementations-7DCE/. Early stopping is implemented to halt training if there's no improvement in the AUC metric after 10 epochs. After training, the model is tested using the test data loader, and performance metrics including ACC, AUC and RMSE.

4.4. Results

This section presents the comparison results of DKT models with separate analyses for forgetting disassembly and decay replacement approaches.

4.4.1. ANOVA Analysis

Table 4 shows the model-specific ANOVA analysis, assessing how much of the variance in model performance can be explained by implementing different decay methods.

The result for each model shows a highly significant intercept effect, emphasising the strong influence of baseline settings specific to each model. Importantly, the heterogeneous response to decay function modifications across models reveals fundamental architectural sensitivities that advance our understanding of forgetting mechanisms in DKT. AKT, in particular, displays a statistically significant impact of the decay method implementations on the model performance (F-statistic = 10.101). The performances of HawkesKT, DKT-Forget, and SimpleKT also show significant effects from using different decay methods, whereas the effects from decay methods applied to RKT and CT-NCM are not statistically significant.

[8] https://edustudio.ai/

[9] Source code and datasets at: https://github.com/alyssa-sha/DKT-Decay-Implementations

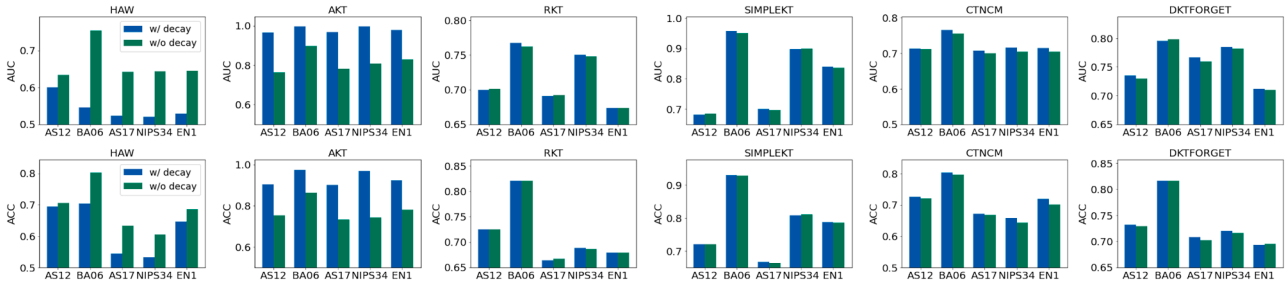


Fig. 5. Performance of DKT models with and without the decay function.

The differential sensitivity suggests that certain architectural designs -particularly those with explicit temporal attention mechanisms like AKT-are more amenable to decay function optimisation, whilst models with built-in temporal regularisation like RKT may already incorporate sufficient forgetting dynamics. These findings provide practical guidance for researchers on which model architectures benefit most from decay function tuning, representing a novel contribution to the field's understanding of model-specific optimisation strategies.

4.4.2. Results of decay disassembly

As depicted in Fig. 5, the ablation results show significant variability across models. The AKT model experienced the most substantial performance degradation, with decreases of 17.13 % in accuracy and 16.87 % in AUC, indicating the critical role of the forgetting mechanisms in this model. The potential reason comes down to that the decay factor in AKT is directly used to adjust the attention score based on the temporal distance between learning events. Its attention mechanism, which explicitly incorporates time dynamics, closely links a student's memory strength to the recency of practice, making the model highly sensitive to time-based decay functions. In contrast, RKT employs a blend of pure attention probabilities, time-decayed attention, and relation-aware attention. This method balances time-based decay with the influence of related knowledge components, reducing its sole reliance on the forgetting curve.

Conversely, the HawkesKT model improved by 10.33 % in accuracy and 22.43 % in AUC after the removal, questioning the plausibility of its original memory decay modelling. One possible explanation is that the model might be overly complex, attempting to account for temporal dynamics more accurately than necessary. The presence of an intricate decay function might introduce noise or cause overfitting, particularly if the time intervals are less critical to knowledge retention than presumed. Additionally, since HawkesKT already implements logarithmic time scaling-reflecting the diminishing impact of time-integrating this with exponential decay might be superfluous and its elimination could simplify the model's predictive accuracy.

Our comparison of DKT models' sensitivity to decay mechanisms suggests that incorporating a decay function within the attention mechanism is a viable strategy. However, beyond the decay curves, it is crucial to thoughtfully determine where to apply the time-based decay function and which parameters it should influence. To prevent overfitting, avoiding overly complex models that include multiple time-scaling processes is essential. Given AKT's success, we recommend that future DKT research using transformers focus on tailoring decay mechanisms to individual differences in forgetting rates. This could be achieved by integrating parameters that adjust the decay based on observed student performance patterns.

4.4.3. Results of decay mechanisms implementations

Fig. 6 illustrates the ACC, AUC, and RMSE differences from the original model across five datasets for three DKT models: HawkesKT, AKT, and RKT, respectively. Each model's performance is measured by the change in different indicators when implementing different decay mechanisms mentioned in step 2 of Methodology section: exponential decay as the benchmark, Sig (Sigmoidal decay), Poly (Polynomial decay), Log (Logarithmic decay), Inv (Inverse decay), and Step decay. Negative values indicate a decrease in ACC and AUC compared to the original model, while positive values denote an improvement. For RMSE, negative values indicate a reduction in error, which reflects better performance.

Our analysis reveals a fundamental problem in current DKT implementations: the assumption that a single decay function can adequately model the complexity of human memory. The observation that inverse and sigmoid decay often outperform exponential decay challenges the traditional reliance on Ebbinghaus's exponential forgetting curve and highlights the need for more nuanced approaches to memory decay modelling. The heterogeneity in model responses to different decay functions suggests that memory types vary significantly in educational contexts. For example, AKT's superior performance with sigmoid decay aligns with Franklin's [14] research on perceptual memory following inverse sigmoidal patterns, whilst semantic memory adheres more closely to Ebbinghaus's exponential decay. This evidence indicates that educational knowledge may involve multiple memory types that require different decay representations. Future KT models are suggested to consider the diversity of memory types-from short-term to long-term and from semantic to perceptual-and tailor decay functions accordingly to better capture the dynamics of human memory.

For the HawkesKT model, inverse decay and sigmoid decay perform better than other methods, with inverse decay showing an average AUC improvement of 0.1289 and ACC of 0.0684. Sigmoid decay follows closely with a 0.1273 increase in AUC and a 0.0682 rise in ACC. With HawkesKT's integration of actual interaction times, inverse decay's emphasis on timing accurately models the rapid decrease in the influence of an event over time, essential for temporal modeling in HawkesKT. While this research does not fully elucidate the reasons behind these patterns, the likely explanation may relate to memory categorisation. As noted in [50], memories from subliminal exposures, such as brief cartoon clips, are typically unconscious and weak, and they exhibit an inverse forgetting pattern. This suggests that the inherent weakness of these memories may actually prioritise them for consolidation, resulting in the observed inverse decay in forgetting. This again challenges the exclusive use of the Ebbinghaus forgetting curve in DKT models, which fails to account for distinct decay patterns of different memory types, and it underscores how minor modifications to memory decay mechanisms can lead to substantial improvements in model performance.

When applied to the RKT model, different decay methods result in relatively stable performance, indicating robustness or a possible insensitivity to these variations. In such cases, adding a decay mechanism might not significantly benefit but would rather increase the complexity and computational cost of training.

To further test the generalisability of modifying the decay mechanism, we implemented the changes on SimpleKT, CT-NCM, and DKT-Forget. The detailed performance comparison for all selected models with different decay mechanisms implemented is shown in Table 5. As SimpleKT and DKT-Forget do not explicitly model forgetting with a

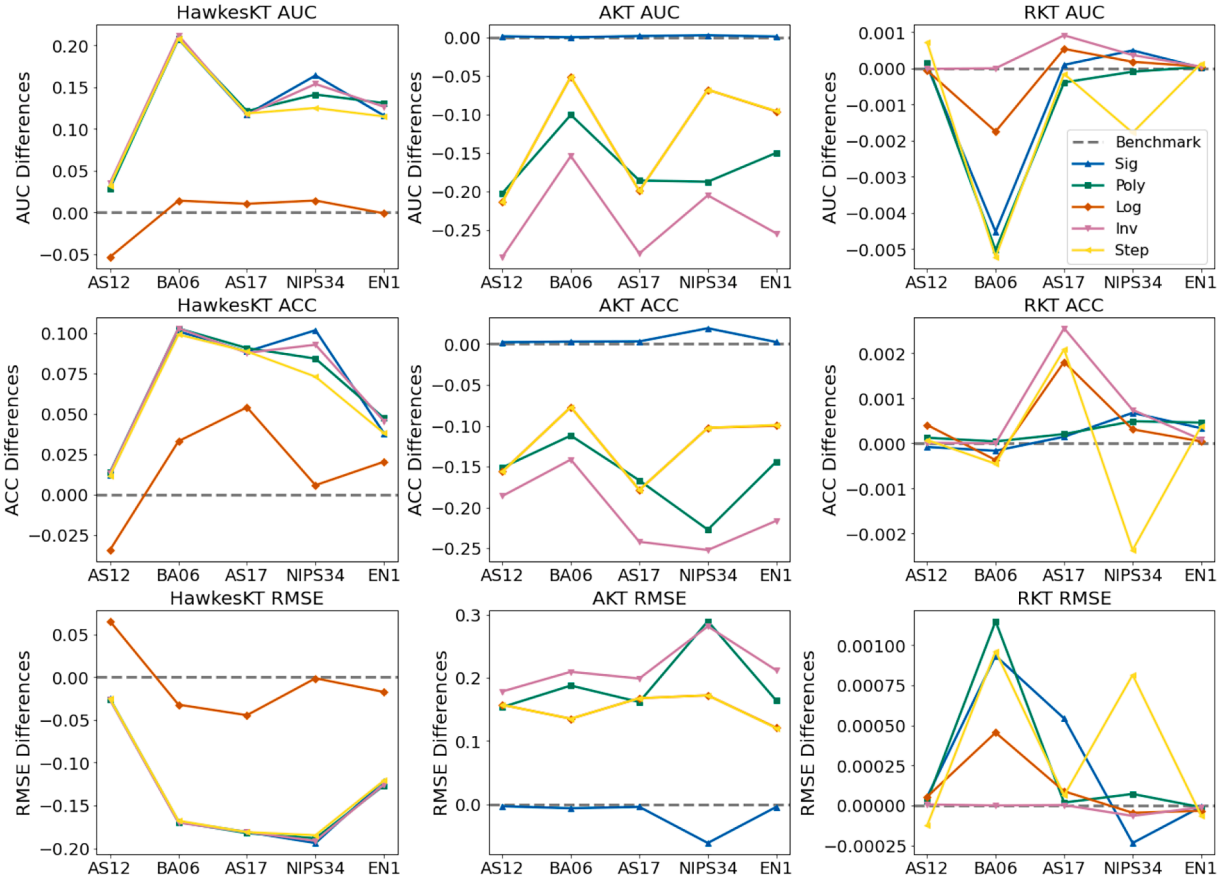


Fig. 6. Performance comparison of DKT models with varying decay mechanisms.

decay curve, we label their original approach as *no decay* and implement exponential decay as an alternative. For other models that already include a decay function using Ebbinghaus forgetting curve, we label the original implementation as *exp decay*.

The best-performing method is highlighted in bold. The "Gain%" represents the maximum improvement in model performance for a specific evaluation metric and dataset achieved by modifying the decay mechanism. While there are instances where the original approach exhibits superior performance, the implementation of decay mechanisms leads to improvements on the majority of datasets compared to the original models, indicating that alternatives to the traditional Ebbinghaus exponential decay has the potential to provide better ways of modelling forgetting in KT.

Our findings expose a critical limitation in current KT datasets: the lack of granular information about memory types and learning contexts that would enable more precise decay function selection. Current datasets provide temporal information but lack cognitive load indicators, content difficulty measures, or memory type classifications that would inform optimal decay mechanism choices. This represents a significant gap in the field that limits our ability to develop theoretically grounded forgetting models. The variability in model performance across datasets further emphasises the context-dependent nature of memory decay. AS12 and AS17's unique attributes, such as spacing of student interactions and longitudinal tracking, affect how forgetting should be modelled. Mathematics-focused datasets (BA06, NIPS34) may require different decay patterns than mixed-subject datasets, suggesting that domain-specific decay functions could provide substantial improvements.

HawkesKT excels in these datasets due to its proficiency in modelling the reinforcement of knowledge through repeated practice. Moreover, features like question bundling in datasets such as EN1 introduce variability in the information available for training, impacting memory retention and the effectiveness of forgetting mechanisms in models like RKT, which showed minimal impact from the removal of forgetting components on the EN1 dataset. To minimise data selection bias in DKT research, we recommend establishing guidelines that require models to be trained on a consistent set of datasets, thereby ensuring more reliable and generalisable performance. Additionally, different data processing logic should be considered when developing benchmark or evaluation tools for KT.

4.4.4. Computational efficiency analysis

To provide a comprehensive evaluation of different decay mechanisms, we analysed the computational cost of implementing various decay functions across the selected DKT models. All experiments were conducted on an NVIDIA DGX Station equipped with 4× Tesla V100-DGXS-32GB GPUs (32GB memory each), NVIDIA driver version 535.230.02, and CUDA 12.2, ensuring consistent and comparable measurements across all model implementations. Table 6 presents the training time measurements for each model with different decay implementations across all selected datasets.

The computational analysis reveals substantial variations across models and datasets, providing crucial insights for practical deployment decisions. RKT demonstrates remarkable efficiency gains on ASSIST12, with sigmoid decay achieving 133.631ms compared to 252.834ms for exponential decay-a 47 % reduction. This improvement likely results from sigmoid's bounded output providing better numerical stability than exponential decay, and its gradient computation being more efficient during backpropagation in RKT's attention mechanism.

SimpleKT shows the largest efficiency variations. On ASSIST17 and EN1, alternative decay functions achieve up to 20× improvements (from 343.918μs to 15.276μs with sigmoid decay on ASSIST17). The

Table 5
Comparative Performance of Decay Methods in Deep Knowledge Tracing Models.

Model	Method	AS12			BA06			AS17			NIPS34			EN1		
		AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
HawkesKT	Original (exp decay)	0.6010	0.6945	0.4729	0.5453	0.7033	0.5438	0.5239	0.5448	0.6553	0.5205	0.5332	0.6683	0.5294	0.6467	0.5763
	Sigmoidal decay	0.6334	0.7065	0.4475	0.7526	0.8043	0.3750	0.6411	0.6332	0.4743	**0.6843**	**0.6348**	**0.4744**	0.6453	0.6847	0.4550
	Polynomial decay	0.6295	0.7062	0.4470	0.7533	0.8061	0.3744	**0.6452**	**0.6354**	**0.4730**	0.6616	0.6173	0.4799	**0.6602**	**0.6940**	**0.4497**
	Log decay	0.5479	0.6602	0.5381	0.5596	0.7363	0.5115	0.5344	0.5987	0.6108	0.5350	0.5390	0.6668	0.5285	0.6670	0.5589
	Inverse decay	**0.6363**	**0.7079**	**0.4463**	**0.7571**	**0.8062**	**0.3736**	0.6410	0.6324	0.4741	0.6742	0.6259	0.4769	0.6559	0.6921	0.4507
	Step decay	0.6334	0.7062	0.4474	0.7529	0.8024	0.3759	0.6424	0.6333	0.4742	0.6455	0.6061	0.4834	0.6443	0.6847	0.4555
	Gain%	5.88	1.92	5.63	38.84	14.63	31.29	23.14	16.63	27.83	31.45	19.07	29.02	24.70	7.31	21.97
AKT	Original (exp decay)	0.9657	0.9046	0.2594	0.9967	0.9752	0.1345	0.9678	0.9001	0.2653	0.9968	0.9699	0.1461	0.9796	0.9246	0.2307
	Sigmoidal decay	**0.9672**	**0.9068**	**0.2567**	**0.9970**	**0.9781**	**0.1285**	**0.9697**	**0.9033**	**0.2615**	**0.9996**	**0.9892**	**0.0852**	**0.9809**	**0.9271**	**0.2268**
	Polynomial decay	0.7631	0.7533	0.4132	0.8961	0.8631	0.3223	0.7819	0.7333	0.4272	0.8093	0.7427	0.4355	0.8296	0.7807	0.3946
	Log decay	0.7520	0.7483	0.4167	0.9456	0.8980	0.2699	0.7683	0.7216	0.4332	0.9288	0.8673	0.3185	0.8835	0.8247	0.3518
	Inverse decay	0.6802	0.7185	0.4379	0.8425	0.8337	0.3441	0.6871	0.6581	0.4642	0.7916	0.7178	0.4275	0.7249	0.7084	0.4426
	Step decay	0.7520	0.7483	0.4167	0.9456	0.8980	0.2699	0.7683	0.7216	0.4332	0.9288	0.8673	0.3185	0.8840	0.8256	0.3511
	Gain%	0.16	0.24	1.07	0.03	0.30	4.47	0.20	0.35	1.43	0.28	1.99	41.68	0.13	0.26	1.67
RKT	Original (exp decay)	0.7000	0.7249	0.4327	**0.7671**	0.8211	**0.3609**	0.6910	0.6637	**0.4624**	0.7502	0.6882	0.4491	0.6738	0.6788	0.4575
	Sigmoidal decay	0.7001	0.7248	0.4327	0.7626	0.8210	0.3619	0.6911	0.6638	0.4629	**0.7507**	0.6889	**0.4489**	0.6738	0.6791	0.4575
	Polynomial decay	**0.7001**	**0.7250**	**0.4327**	0.7621	**0.8212**	0.3621	0.6906	0.6639	0.4624	0.7501	0.6887	0.4492	0.6738	0.6793	0.4575
	Log decay	0.6999	0.7253	0.4327	0.7653	0.8208	0.3614	0.6915	0.6655	0.4625	0.7503	0.6885	0.4491	0.6738	0.6789	0.4575
	Inverse decay	0.7000	0.7249	0.4327	**0.7671**	0.8211	**0.3609**	**0.6919**	**0.6662**	0.4624	0.7505	**0.6890**	0.4490	0.6738	0.6789	0.4575
	Step decay	0.7007	0.7250	0.4325	0.7619	0.8207	0.3619	0.6909	0.6657	0.4625	0.7484	0.6859	0.4499	**0.6739**	**0.6792**	**0.4575**
	Gain%	0.10	0.06	0.03	-	0.01	-	0.13	0.38	-	0.07	0.11	0.05	0.02	0.07	0.01
SimpleKT	Original (no decay)	**0.6848**	0.7208	0.4368	0.9509	0.9282	0.2398	0.6966	0.6641	0.4606	0.8990	0.8103	0.3588	0.8372	0.7854	0.3790
	exp decay	0.6821	0.7206	**0.4366**	0.9571	**0.9298**	**0.2334**	0.7002	0.6673	0.4584	0.8971	0.8072	0.3605	**0.8406**	**0.7878**	0.3759
	Sigmoidal decay	0.6821	0.7201	0.4371	**0.9573**	0.9296	0.2335	**0.7024**	**0.6690**	**0.4579**	**0.9015**	0.8117	**0.3565**	0.8405	0.7873	**0.3758**
	Polynomial decay	0.6801	0.7208	0.4371	0.9558	0.9289	0.2349	0.6935	0.6617	0.4607	0.8994	0.8097	0.3584	0.8336	0.7818	0.3816
	Log decay	0.6825	0.7209	0.4371	0.9421	0.9159	0.2548	0.6918	0.6635	0.4608	0.8997	**0.8123**	0.3586	0.8399	0.7833	0.3800
	Inverse decay	0.6531	0.7112	0.4437	0.9314	0.9163	0.2613	0.6507	0.6406	0.4716	0.7833	0.7050	0.4376	0.7449	0.7287	0.4267
	Step decay	0.6825	**0.7209**	0.4371	0.9421	0.9187	0.2509	0.6918	0.6635	0.4608	0.9008	0.8033	0.3632	0.8399	0.7833	0.3800
	Gain%	-	0.01	0.03	0.68	0.17	2.66	0.83	0.73	0.59	0.28	0.25	0.64	0.41	0.31	0.85
CT-NCM	Original (exp decay)	0.7132	0.7257	0.4304	**0.7650**	**0.8044**	**0.3825**	0.7070	0.6722	0.4569	0.7157	0.6578	0.4632	**0.7151**	0.7196	0.4360
	Sigmoidal decay	0.7139	0.7268	0.4302	0.7062	0.6740	0.4572	**0.7662**	**0.8066**	**0.3733**	0.7164	0.6582	0.4631	0.7148	**0.7207**	**0.4357**
	Polynomial decay	0.7140	0.7270	0.4301	0.7062	0.6738	0.4572	0.7662	0.8066	0.3733	**0.7165**	**0.6594**	**0.4628**	0.7148	0.7207	0.4357
	Log decay	**0.7141**	**0.7270**	**0.4301**	0.7062	0.6738	0.4573	0.7662	0.8066	0.3733	0.7153	0.6568	0.4638	0.7148	0.7207	0.4357
	Inverse decay	0.7140	0.7268	0.4302	0.7062	0.6741	0.4573	0.7662	0.8066	0.3733	0.7160	0.6563	0.4639	0.7148	0.7207	0.4357
	Step decay	0.7139	0.7268	0.4302	0.7062	0.6740	0.4572	0.7662	0.8066	0.3733	0.7153	0.6580	0.4633	0.7148	0.7207	0.4357
	Gain%	0.13	0.17	0.07	-	-	-	8.37	19.99	18.30	0.11	0.25	0.08	-	0.15	0.06
DKT-Forget	Original (no decay)	0.7293	0.7293	0.4266	0.7978	0.8158	0.3615	0.7591	0.7018	0.4399	0.7820	0.7166	0.4338	0.7101	0.6950	0.4475
	exp decay	0.7348	0.7322	0.4247	0.7951	0.8158	0.3603	**0.7666**	0.7083	**0.4362**	**0.7850**	**0.7197**	**0.4326**	0.7118	0.6934	0.4473
	Sigmoidal decay	**0.7355**	0.7323	**0.4245**	0.8008	0.8122	0.3607	0.7646	0.7071	0.4373	0.7835	0.7185	0.4333	**0.7157**	**0.6952**	**0.4469**
	Polynomial decay	0.7037	0.7176	0.4408	0.7514	0.8058	0.3730	0.7320	0.6837	0.4516	0.7771	0.7142	0.4365	0.7074	0.6947	0.4482
	Log decay	0.7339	0.7318	0.4253	0.7574	0.8070	0.3703	0.7588	0.7014	0.4402	0.7826	0.7176	0.4336	0.7132	0.6947	0.4474
	Inverse decay	0.7341	0.7329	0.4247	**0.8015**	0.8151	0.3595	0.7643	0.7058	0.4373	0.7839	0.7182	0.4330	0.7118	0.6931	0.4474
	Step decay	0.7347	**0.7329**	0.4245	0.8004	**0.8189**	**0.3590**	0.7654	**0.7084**	0.4368	0.7847	0.7194	0.4327	0.7119	0.6948	0.4473
	Gain%	0.85	0.49	0.50	0.46	0.38	0.69	0.98	0.94	0.84	0.39	0.43	0.27	0.78	0.03	0.13

original SimpleKT without decay is substantially slower, indicating that any decay mechanism improves efficiency by constraining attention computations.

HawkesKT consistently improves with polynomial, sigmoid, and step decay functions, achieving 25–30 % reductions across datasets. These simpler decay computations reduce the overhead of temporal cross-effects calculations. Importantly, these efficiency gains occur alongside performance improvements shown in (Table 6).

AKT and CT-NCM show stable computational profiles. AKT's transformer attention mechanism shows variations under 25 % across different decay functions, as the decay computation represents a small fraction of the total attention calculation cost. CT-NCM shows minimal differences (under 10 %), suggesting its architecture is less dependent on decay function choice.

Dataset size affects both computational and performance patterns. Larger datasets (ASSIST12 with 2.5M interactions) show more pronounced efficiency gains and performance variations. Smaller datasets (NIST34 with 1.4M interactions) show less differences, suggesting decay function selection becomes more critical as dataset size and learning period increase.

In summary, the right choice of decay function enhances not only model performance but also computational efficiency. Sigmoid and polynomial decay functions often provide computational benefits alongside performance improvements, rather than requiring trade-offs between efficiency and accuracy. Based on our findings, we encourage researchers to try sigmoid decay for attention-based models (AKT, RKT) as it brings numerical stability and efficient gradient computation, and to try polynomial or step decay for temporal models (HawkesKT) as they

Table 6
Computational Cost Comparison Across DKT Models and Datasets.

Model	Operations	ASSIST12 (ms)	ASSIST17 (μs)	BA06 (μs)	NIPS34 (μs)	EN1 (μs)
HawkesKT	Original (Exp decay)	250.931	14.78	17.683	267.761	298.25
	Sigmoid decay	**187.567**	17.205	18.729	**185.39**	**16.632**
	Log decay	224.268	**15.543**	**17.912**	230.713	15.759
	Step decay	**186.857**	16.116	16.957	**193.163**	16.305
	Inverse decay	191.695	17.592	17.392	191.704	16.478
	Polynomial decay	**182.602**	15.479	17.38	197.582	**15.535**
AKT	Original (Exp decay)	234.263	260.833	244.87	225.502	244.164
	Sigmoid decay	285.046	265.201	284.264	297.353	271.412
	Log decay	290.237	284.055	276.886	**243.865**	278.594
	Inverse decay	**241.262**	292.211	287.918	289.418	282.375
	Polynomial decay	272.994	**263.856**	268.771	263.132	285.583
	Step decay	284.721	297.11	**258.944**	276.979	280.84
RKT	Original (Exp decay)	252.834	25.59	57.663	252.211	26.766
	Sigmoid decay	**133.631**	**24.54**	61.691	232.975	**25.552**
	Log decay	229.683	**22.328**	58.648	229.959	25.711
	Polynomial decay	253.16	23.323	60.861	242.102	**24.997**
	Step decay	235.635	252.456	**58.138**	**218.645**	25.191
	Inverse decay	239.766	23.115	**55.884**	239.644	**24.49**
SimpleKT	Original	348.492	343.918	357.985	361.39	347.66
	Exp decay	**242.729**	303.571	**16.387**	247.567	**14.893**
	Sigmoid decay	**216.894**	**15.276**	16.303	249.63	15.198
	Log decay	238.161	16.009	15.969	**234.656**	303.648
	Inverse decay	262.214	15.937	17.266	261.682	14.965
	Polynomial decay	255.908	**15.281**	16.286	257.444	15.506
	Step decay	271.828	15.415	**15.82**	275.711	**14.282**
CT-NCM	Original (Exp decay)	5.912	6.364	6.518	6.392	6.198
	Sigmoid decay	6.233	6.555	**6.156**	**6.019**	6.237
	Polynomial decay	**5.95**	6.65	6.451	**5.873**	6.599
	Log decay	6.067	6.565	6.351	6.422	6.293
	Inverse decay	6.278	**6.269**	6.636	6.044	6.768
	Step decay	6.201	6.549	6.365	6.231	6.241
DKT-Forget	Original	2.194	28.539	76.639	260.678	403.961
	Exp decay	2.168	29.742	79.73	258.156	**363.986**
	Sigmoid decay	2.183	29.238	77.923	**22.755**	342.946
	Polynomial decay	**2.143**	**28.361**	77.388	251.298	398.937
	Log decay	2.187	28.927	**75.478**	270.908	381.768
	Inverse decay	**2.16**	29.174	81.675	248.248	388.647
	Step decay	2.176	**27.655**	75.491	267.109	405.233

Note: Training times are measured in milliseconds (ms) for DKT-Forget on ASSIST12 dataset, and microseconds (μs) for all other measurements. Bold values indicate the best (lowest) computational cost for each model-dataset combination.

bring both performance and efficiency gains. However, performance and efficiency are not the only variables to consider-dataset characteristics also influence optimal decay function selection, though identifying specific dataset features that predict decay function effectiveness remains an open research question.

5. Conclusion

This paper identifies a fundamental problem in Deep Knowledge Tracing: the overreliance on Ebbinghaus's exponential decay function without consideration of others. Through comparative analysis of decay functions across six SOTA DKT models and five commonly-used KT datasets, we demonstrate that alternative decay mechanisms can outperform the traditional approach in many scenarios, with systematic evidence that different educational contexts require correspondingly diverse decay functions rather than universal exponential decay. Our primary contribution is raising awareness of this limitation in a popular field, most of which adopt the same decay assumption without theoretical justification. We have systematically identified shortcomings in existing implementations and provided evidence that the one-size-fits-all approach to memory decay is inadequate. Through establishing linkages between psychological theories of memory and DKT implementations, we show that different types of memories require different implementations of decay functions. Our ablation studies reveal that the effectiveness of forgetting mechanisms varies across learning contexts and model architectures, providing practical insights for educational system design.

Our computational efficiency analysis demonstrates that implementing alternative decay functions is practically feasible, with some alternatives providing both performance and computational benefits. The standardised training pipeline we implemented (available at https://anonymous.4open.science/r/DKT-Decay-Implementations-7DCE/) enables systematic comparison of different decay functions across multiple DKT models, facilitating future research in this area.

Building on these findings, we summarise several promising research directions. An extension would be integrating interference theory with decay functions to model how competing knowledge items affect retention. Further investigation could explore the relationship between content difficulty and optimal decay patterns, potentially leading to more accurate modelling approaches. Additionally, developing adaptive decay mechanisms that adjust based on content type and learning context could enhance the precision of DKT. While our study focused primarily on decay functions, future work examining complete dropout layers and other architectural variations could provide complementary insights into modelling forgetting in KT tasks.

Limitations of our study include the constraint of existing KT datasets, which do not clearly consider memory decay and interference effects thus lack the granular information needed for optimal decay function selection. Future work should prioritise the development of more comprehensive educational datasets that capture the different forgetting mechanisms.

CRediT authorship contribution statement

Alyssa Sha: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zhenhui Gu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Data curation; **Bernardo Pereira Nunes:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization.

Data availability

The data and code are shared using public git repository. Link is provided in the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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